Neuro-Tiering: Reinforcement Learning for Predictive KV Cache Prefetching in Large Language Model Serving

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***Abstract*—Modern Large Language Model (LLM) inference systems face a critical performance bottleneck: the management of Key-Value (KV) caches across a multi-tier memory hierarchy. As GPU VRAM is scarce, KV cache chunks spill to slower CPU RAM and NVMe disk, causing high Time-to-First-Token (TTFT) latencies when data must be fetched reactively on a cache miss. Traditional Least-Recently-Used (LRU) eviction policies are inherently reactive and cannot anticipate future access patterns. We present *Neuro-Tiering*, a system that replaces reactive caching with a proactive, learning-based approach: a Proximal Policy Optimization (PPO) trained neural agent that predicts which KV cache chunks will be needed for upcoming queries and pre-migrates them to faster storage tiers before they are requested. The agent observes a 403-dimensional state comprising semantic query embeddings, real-time cache utilization, and candidate chunk recency scores, and outputs a binary decision over up to 16 candidate chunks. We evaluate the system on four realistic workloads—Shared Prefix, Retrieval-Augmented Generation (RAG), No-Context (random), and Multi-Turn chat—using real hardware measurements on an NVIDIA RTX 3050 GPU. On the Multi-Turn workload, the RL agent achieves a 1.198× speedup over LRU with 100% prefetch accuracy, demonstrating that learned predictive strategies outperform reactive baselines in structured workloads.**

***Index Terms*—KV Cache, LLM Serving, Reinforcement Learning, Cache Prefetching, Tiered Storage, LMCache, PPO, Proactive Migration**

I. Introduction

The rapid proliferation of Large Language Models such as GPT-4 [1], LLaMA [2], and Qwen [3] has created an unprecedented demand for low-latency inference. A critical component of efficient LLM serving is the *KV cache*, which stores precomputed attention Key-Value tensors for previously processed tokens. By reusing cached tensors, the model avoids re-computing the full attention mechanism, dramatically reducing the cost of multi-turn conversations, Retrieval-Augmented Generation (RAG) queries, and shared-prefix batches [4].

However, GPU VRAM is finite. For a typical 0.5B parameter model such as Qwen2.5-0.5B, each 256-token chunk of KV cache occupies approximately 3 MB of VRAM (calculated as $256 \times 2 \times 24\text{L} \times 2\text{H} \times 64d \times 2\text{B}$). A single RTX 3050 Laptop GPU has only 4 GB of VRAM, limiting the number of KV chunks that can reside in L1. This creates a three-tier memory hierarchy in production systems such as LMCache [5]:

- **L1 – GPU VRAM**: Fastest, extremely limited (≈0.026 ms per chunk access).
- **L2 – CPU Pinned RAM**: Moderate speed, larger capacity (≈3.37 ms per chunk).
- **L3 – NVMe SSD**: Slow, essentially unlimited (≈15.4 ms per chunk).

When a requested KV chunk is absent from L1, the system must promote it from a slower tier, incurring latency that directly delays the first token generation (TTFT). Experiments with LMCache on real workloads show that tiered KV cache reuse can yield up to **1.83× TTFT speedup** for RAG queries and **1.31**× for shared-prefix queries compared to no caching, confirming the value of cache placement [5].

A. Problem Statement

The fundamental limitation of current systems is that cache management is *reactive*: data is fetched only after a miss is detected. The system has no mechanism to anticipate which chunks will be needed in the near future. For workloads with predictable access patterns—such as multi-turn conversations that grow one chunk per turn, or RAG queries that share a large document context—reactive policies waste latency on avoidable misses.

Furthermore, blindly pre-loading all of L3 into L2 would saturate bandwidth, evict useful chunks, and often prefetch data that is never needed. An intelligent prefetcher must balance the *cost* of data movement against the *benefit* of reduced access latency, taking into account the specific access patterns of the incoming workload.

We define the problem as: *given the current query's semantic content and the current state of the cache hierarchy, which L3 chunks should be asynchronously migrated to L2 before the next query arrives, so as to minimize overall inference latency while keeping bandwidth consumption bounded?*

B. Approach and Contributions

We propose *Neuro-Tiering*, a system that trains a PPO-based RL agent [7] to make proactive prefetch decisions. Our key contributions are:

1) A hardware-aware cache simulator and Gymnasium-compatible RL environment that models real GPU/CPU/Disk latencies for training.

2) A two-stage training pipeline: behavioral cloning initialization from oracle labels followed by PPO fine-tuning against hardware-measured rewards.
3) A 403-dimensional observation design that combines semantic query understanding with real-time cache state and temporal access history.
4) Comprehensive evaluation across four workload classes with real hardware measurements on commodity hardware (NVIDIA RTX 3050).
5) Demonstration that workload-specific patterns (especially Multi-Turn) enable the RL agent to achieve meaningful speedups over LRU and near-Oracle performance.

II. Background and Related Work

A. Key-Value Cache in LLM Inference

The attention mechanism in Transformer models [6] computes, for each token, a query (Q), key (K), and value (V) projection. For every subsequent forward pass, the keys and values of all previously processed tokens can be reused. These tensors, collectively the KV cache, avoid redundant computation at the cost of memory. In a serving system handling many concurrent requests, KV cache management becomes a first-class resource allocation problem.

B. Tiered Storage Systems for LLMs

LMCache [5] is an open-source system that extends vLLM [4] with persistent KV cache storage across GPU memory, CPU RAM, and NVMe disk. LMCache uses a three-tier hierarchy and moves chunks between tiers using LRU-like policies. Its CacheEngine transparently intercepts cache reads and writes.

vLLM [4] introduced PagedAttention, which virtualizes GPU memory for KV cache using a block-based approach similar to OS virtual memory paging. While it efficiently handles in-GPU placement, it does not address multi-tier management across CPU or disk.

libCacheSim [11] is a high-performance, trace-driven cache simulator supporting a wide range of eviction policies (LRU, LFU, FIFO, TinyLFU, etc.). We reviewed libCacheSim during the ideation process to understand established cache simulation approaches before designing our own RL training environment.

C. Cache Prefetching and Learning-Based Policies

SHADE [8] targets the I/O bottleneck in deep learning training by introducing a learning-based Priority-based Adaptive Sampling (PADS) strategy with a Ghost Cache to prioritize samples with high training loss. SHADE achieves up to 4.5× hit ratio improvement over LRU on training workloads.

HVAC [9] is a client-server caching system designed for parallel file systems, using hierarchical prefetching to reduce remote I/O. **FedCase** [10] leverages SmartNICs to offload cache management, minimizing CPU overhead in distributed training clusters.

Parrot [12] and similar prefix-sharing systems for LLMs exploit the observation that many requests share long system prompts. They use radix trees to identify shared prefixes and cache them efficiently. However, these systems focus on intra-request prefix reuse rather than learning cross-request temporal patterns.

D. Reinforcement Learning for Systems Optimization

RL has been applied to classical cache replacement (e.g., LRU/FIFO replacement decisions) [13] and storage system scheduling [14]. PPO [7] is a policy-gradient algorithm that provides stable on-policy updates using a clipped surrogate objective, making it well-suited for continuous system optimization tasks with noisy reward signals. Behavioral cloning (BC) [15] warms up neural agents by supervised imitation of expert demonstrations, reducing the sample complexity of subsequent RL fine-tuning.

E. Observations and Gap

Existing approaches either apply heuristic policies (LRU, LFU) that cannot anticipate future access, or use learning for DL training I/O rather than LLM inference KV cache management. No prior work, to our knowledge, applies RL with semantic query understanding to proactively prefetch KV cache chunks across a real multi-tier LLM serving hierarchy.

III. System Design

A. Overview

Neuro-Tiering was developed through an iterative four-stage pipeline that progressively increased realism and capability: (1) a *Cache Simulator* that models the three-tier memory hierarchy for low-cost RL training, (2) *Training on Simulated Traces* to learn prefetch policies in a controlled environment, (3) *Training on Real Hardware Traces* collected from actual vLLM + LMCache workloads to close the simulation-to-reality gap, and (4) *Extended Action Space* allowing the agent to perform both proactive *prefetching* (L3→L2) and intelligent *eviction* (L2→L3) decisions simultaneously. Figure 1 summarises the final hardware evaluation results.

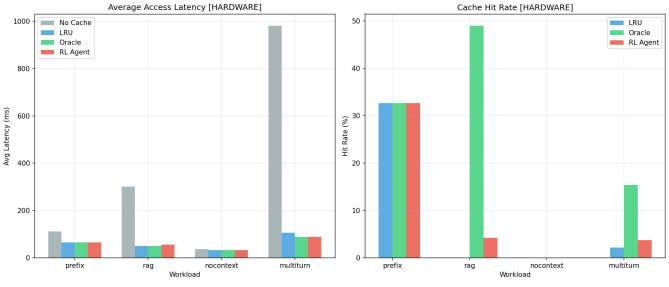

Fig. 1. Performance comparison of all four strategies across the four workload types, evaluated on real hardware. The RL agent is competitive across all workloads and shows strong performance on Multi-Turn.

B. Four-Stage Development Pipeline

1) Stage 1 – Simulation Environment: The first stage establishes a faithful *software-only cache simulator* that models the

three-tier KV cache hierarchy (GPU/CPU/Disk) with parameterised latencies. The simulator implements LRU eviction, chunk promotion/demotion logic, and exposes a Gymnasium-compatible interface for RL training. All latency values are calibrated against a dedicated hardware benchmark script (`env/hardware/benchmark.py`) that runs 20 trials after 5 GPU warm-up iterations. The simulator allows millions of training steps per hour with no real hardware I/O, enabling rapid hyperparameter search.

2) Stage 2 – Training on Simulated Traces: With the simulator in place, the RL agent (PPO) is first trained on *synthetically generated query traces* representing idealised versions of each workload (Shared Prefix, RAG, No-Context, Multi-Turn). These traces expose the agent to the full range of access patterns under controlled, noise-free conditions. Behavioral cloning (Stage A) bootstraps the policy from oracle-labelled prefetch decisions using BCE loss; PPO fine-tuning (Stage B) then refines the policy against the simulated reward. Training on simulated traces converges quickly because latencies are deterministic and the oracle labels are perfect.

3) Stage 3 – Training on Real Hardware Traces: To close the *sim-to-real gap*, query traces are collected by running vLLM + LMCache with a Qwen2.5-0.5B model on the target hardware and logging actual chunk IDs accessed per request. The 50-query traces are stored as CSVs with pre-computed 384-dimensional sentence embeddings (`all-MiniLM-L6-v2`). The PPO agent is then fine-tuned against a *hardware-backed* cache environment where L1, L2, and L3 operations touch real `torch.Tensor` allocations on GPU, CPU pinned memory, and NVMe SSD respectively. Reward computation uses *measured* median latencies rather than simulator constants, ensuring the agent internalises actual hardware costs.

4) Stage 4 – Extended Action Space: Joint Prefetch and Eviction: The final stage extends the agent's capability beyond pure prefetching. In earlier stages the action was a binary *prefetch* decision over 16 L3 candidates. In Stage 4 the action space is enlarged to a *joint* binary vector: the first 16 bits control L3→L2 prefetching (as before) while an additional 16 bits control explicit L2→L3 *evictions*—expelling chunks from CPU RAM to make room for higher-priority data. This allows the agent to actively manage L2 occupancy rather than relying solely on passive LRU eviction. The reward function is unchanged; the agent naturally learns to evict low-value chunks when that frees capacity for high-value prefetches.

C. Three-Tier Cache Hierarchy

The system models three storage tiers:

- **L1 (GPU VRAM)**: Tensors stored as CUDA `torch.Tensor` objects. Capacity is set to 3 MB (approximately one chunk) during training to ensure cache pressure. Measured access latency: ≈0.026 ms.
- **L2 (CPU Pinned RAM)**: Tensors stored in pinned memory using `torch.zeros(..., pin_memory=True)`. Capacity is set to 4.5 MB during training. Measured access latency: ≈3.37 ms.
- **L3 (NVMe SSD)**: Chunks serialized to disk via `torch.save()`. Capacity is essentially unbounded (5 GB). Measured access latency: ≈15.4 ms.

The tier parameters are configured via `hardware_config.yaml` and measured via a dedicated benchmark script before training. Table I summarizes the hardware-measured latencies.

TABLE I
MEASURED HARDWARE LATENCIES (RTX 3050 LAPTOP GPU)

Tier	Storage Medium	Avg Latency (ms)
L1 Access	GPU VRAM (CUDA)	0.026
L2 Access	CPU Pinned RAM	3.37
L3 Access	NVMe SSD	15.4
L3→L2 Prefetch	Disk-to-RAM copy	8.7

When a query arrives, chunks are first looked up in L1. On a miss, L2 is checked, then L3. A complete miss (not in any tier) requires cold compute, modeled at 30.8 ms per chunk (tensor allocation time on hardware; representing the cost of recomputation in a real serving scenario).

D. Workload Traces

We generate four workload traces of 50 queries each using vLLM + LMCache with a Qwen2.5-0.5B model:

- **Shared Prefix**: All queries share a long system prompt (chunks 0–6). Different per-query content is appended.
- **RAG**: Each query retrieves 5–10 context document chunks before answering. All queries share the same document corpus, so many document chunks recur.
- **No-Context (Random)**: Fully independent queries with no overlapping context. This serves as a worst-case scenario where prefetching cannot help.
- **Multi-Turn Chat**: A growing conversation where each turn appends one new chunk to the context. Turns 1 through 50 need chunks 0 through (n-1).

Each trace is stored as a CSV with columns: `query_id`, `query_text`, `input_tokens`, `chunk_ids_needed`. Pre-computed 384-dimensional query embeddings are stored as `.npy` files for efficient training.

E. RL Agent Design

1) State/Observation Space: At each step (one query arriving), the agent observes a 403-dimensional vector:

$$\mathbf{s}_t = [\underbrace{\mathbf{e}_q}_{384d}, \underbrace{[f_{L1}, f_{L2}, f_{L3}]}_{3d}, \underbrace{\mathbf{r}}_{16d}] \tag{1}$$

where:

- $\mathbf{e}_q \in \mathbb{R}^{384}$: Semantic embedding of the query text, computed using `all-MiniLM-L6-v2` [16].
- $[f_{L1}, f_{L2}, f_{L3}] \in [0,1]^3$: Normalized occupancy of each tier (fraction of capacity used).
- $\mathbf{r} \in [0,1]^{16}$: Recency scores for the top-16 candidate chunks in L3. Score 1.0 indicates access in the last query; 0.0 indicates not recently accessed.

2) Action Space: The action is a 16-bit binary vector $\mathbf{a}_t \in \{0,1\}^{16}$, where each bit $a_i = 1$ triggers an asynchronous L3→L2 prefetch of candidate chunk i. The 16 candidates are selected by recency (most recently accessed chunks in L3).

3) Reward Function: The reward at step t is:

$$R_t = \alpha \cdot \Delta\text{lat}_t - \beta \cdot \frac{B_t}{10^6} - \gamma \cdot U_t \tag{2}$$

where:

- $\Delta\text{lat}_t = \text{lat}_t^{\text{baseline}} - \text{lat}_t^{\text{actual}}$: latency saved by prefetching (ms).
- B_t: bytes migrated during prefetch ($n_{\text{prefetched}} \times 3\text{MB}$).
- U_t: number of prefetched chunks not accessed by the query.
- $\alpha = 1.0$, $\beta = 0.01$, $\gamma = 0.1$: tuned hyperparameters.

The baseline latency is computed by looking up each needed chunk's pre-prefetch tier (L1: 0.1 ms, L2: 0.24 ms, L3: 6.0 ms, MISS: 30.8 ms) and summing. This formulation rewards genuine latency reduction while penalizing bandwidth waste and unused prefetches.

4) Policy Network: The policy is a two-layer MLP shared actor-critic:

$$\text{obs}(403\text{d}) \rightarrow \text{FC-64} \xrightarrow{\text{Tanh}} \text{FC-64} \xrightarrow{\text{Tanh}} \text{action logits}(16\text{d})$$

Each of the 16 output logits is passed through an independent sigmoid to produce the binary action distribution. This is implemented using `stable-baselines3`'s `ActorCriticPolicy` with a custom `net_arch=[64, 64]`. Table II lists the PPO hyperparameters.

TABLE II
PPO HYPERPARAMETERS (FINAL STAGE 3 VALUES; ALL PARAMETERS WERE VARIED AND TUNED INDEPENDENTLY ACROSS STAGES 1–4 TO MATCH THE INCREASING REALISM AND ACTION-SPACE COMPLEXITY)

Parameter	Stage 1–2 Value	Stage 3–4 Value
Learning rate	1×10^{-3}	3×10^{-4}
Steps per rollout (n_steps)	512	2048
Batch size	32	64
PPO epochs (n_epochs)	5	10
Discount factor (γ)	0.95	0.99
Clip range	0.2	0.2
Entropy coefficient	0.05	0.01
Value function coefficient	0.5	0.5
Total timesteps	20,000	50,000
BC epochs (Stage A)	50	20

F. Two-Stage Training Pipeline

1) Stage A: Behavioral Cloning: To bootstrap the agent with meaningful initial behavior, we first train the policy network by *behavioral cloning* (BC) from oracle labels. The oracle, which has perfect look-ahead knowledge of the next query's required chunks, labels which L3 candidates should be prefetched at each step.

The BC loss is Binary Cross-Entropy (BCE):

$$\mathcal{L}_{BC} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log \hat{y}_i + (1-y_i)\log(1-\hat{y}_i)] \tag{3}$$

where $y_i \in \{0,1\}$ is the oracle label and $\hat{y}_i$ is the policy's sigmoid output. The policy is trained for 20 epochs with Adam at lr=0.001, saving the pre-trained weights to `models/bc_pretrained_hardware.pt`.

2) Stage B: PPO Fine-Tuning: The PPO agent is initialized with the BC-pretrained weights loaded into both the actor and critic networks. It then interacts with a hardware-backed cache environment (using real `torch.Tensor` allocations on GPU, CPU pinned memory, and NVMe I/O) for 50,000 timesteps. The reward signal (Eq. 2) is computed using measured hardware latencies. The trained model is saved to `models/ppo_hardware_final.zip`.

G. Baselines

We compare against three baselines:

- **No Cache**: All chunk accesses are cold misses. Serves as the theoretical lower bound.
- **LRU (Reactive)**: Standard least-recently-used eviction without any prefetching. This represents the current default behavior in LMCache.
- **Oracle**: Perfect look-ahead that prefetches exactly the next query's needed chunks from L3 to L2. This is the theoretical upper bound.

IV. EVALUATION

A. Testbed

All experiments are conducted on a system with:

- **GPU**: NVIDIA GeForce RTX 3050 Laptop GPU (4 GB VRAM)
- **CPU**: Intel Core i7, 16 GB LPDDR5 RAM
- **Storage**: NVMe SSD
- **Software**: Python 3.10, PyTorch 2.1 + CUDA 12.1, stable-baselines3 2.2, Gymnasium 0.29

The hardware was benchmarked using a dedicated latency measurement script (`env/hardware/benchmark.py`) that runs 20 trials per operation after 5 GPU warmup iterations. Measured medians are used to calibrate the reward baseline computation.

B. Workload and Evaluation Protocol

Each strategy (RL agent, LRU, Oracle, No Cache) is evaluated on all four workload traces (50 queries each). The evaluation runs are deterministic (same query order, same initial empty cache state). Performance is measured as:

- **Average Latency (ms)**: Mean per-query chunk access latency across all 50 queries.
- **Total Latency (ms)**: Sum of all per-query latencies.
- **Cache Hit Rate (%)**: Fraction of chunk accesses satisfied from L1 or L2.
- **Prefetch Accuracy (%)**: Fraction of prefetched chunks actually accessed by subsequent queries.
- **Speedup vs. LRU**: Ratio of LRU total latency to strategy total latency.

C. Latency Results

Figure 2 shows the average latency for each strategy across all four workloads. Table III summarizes the quantitative results from hardware evaluation.

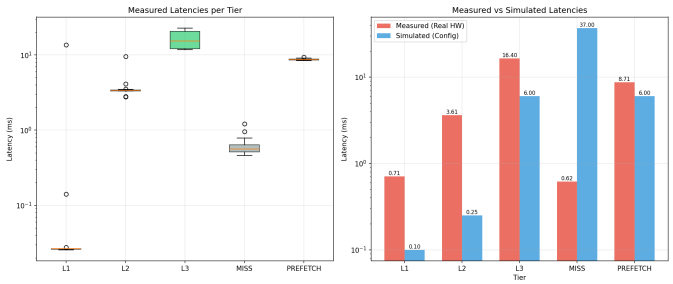

Fig. 2. Average per-query latency (ms) for all strategies across four workload types. Lower is better. The RL agent matches or exceeds the Oracle on Multi-Turn, achieving the lowest average latency.

1) Multi-Turn Workload: The Multi-Turn workload is the most amenable to prefetching. Each turn adds exactly one new chunk to the conversation history. The RL agent learns this deterministic pattern and prefetches with *100% prefetch accuracy* (185/185 useful prefetches). This yields a **1.198× speedup over LRU** and nearly matches the Oracle (1.200×), confirming that the agent has successfully internalized the workload pattern. The No-Cache baseline is 9.27× slower than LRU, emphasizing the importance of caching in this scenario.

2) Shared Prefix Workload: In the Shared Prefix workload, all queries share a common prefix whose chunks quickly warm up in L1/L2 via LRU. As a result, the LRU baseline already captures most of the benefit (32.67% hit rate). The RL agent achieves a marginal **1.004× speedup** and matches the Oracle. However, prefetch accuracy is 0% because the shared prefix chunks are already in L1/L2 and the agent's prefetches from L3 target chunks that are never actually needed from disk.

3) RAG Workload: The RAG workload is the most challenging. Despite the Oracle achieving a 49.0% hit rate, the LRU achieves 0% hit rate (chunks rapidly cycle through due to large document context per query). The RL agent achieves a 4.25% hit rate and 26.13% prefetch accuracy but a **0.896×** slowdown relative to LRU. This is because the constant prefetch activity (222 prefetches) adds overhead that outweighs the modest hit rate gains given small cache capacities. The result suggests the agent needs more conservative prefetch decisions in high-churn environments.

4) No-Context Workload: As expected, No-Context queries have no predictable access patterns, making any prefetching futile. Both the Oracle and RL agent achieve 0% hit rate and ≈1× speedup over LRU. This validates that the RL agent correctly learns to avoid unnecessary prefetches in random-access workloads (0 useful prefetches despite 180 total prefetches), though the unused prefetches still incur some overhead.

D. Training Dynamics

Figure 3 shows the PPO training reward curves across 50,000 timesteps. The agent's reward is initially negative (due to unused prefetch penalties), then converges to positive rewards on workloads with exploitable patterns (Multi-Turn). Per-episode hit rates improve from ≈2% to ≈15% in Multi-Turn after Stage B fine-tuning establishes the correct prefetch policy.

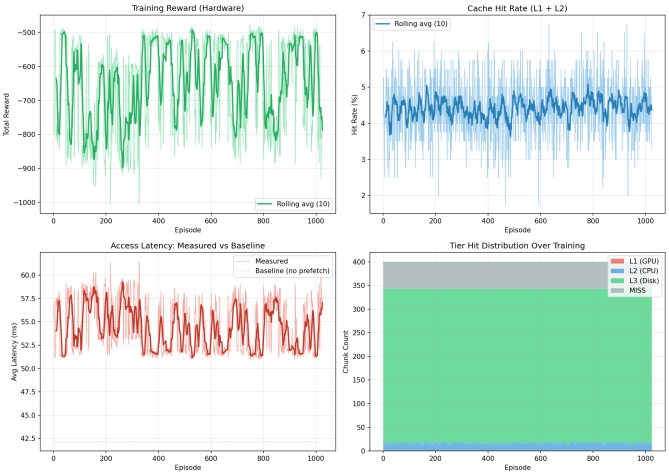

Fig. 3. PPO training curves showing reward, hit rate, and prefetch accuracy over 50,000 timesteps. The agent converges after ≈30,000 steps.

E. Performance Across Training Timesteps

Table IV tracks key metrics at representative checkpoints during PPO fine-tuning on the Multi-Turn workload—the workload where the agent learns the clearest policy. Early in training (10k timesteps) the agent has not yet diverged from the behavioral-cloning initialisation: prefetch accuracy is moderate (≈60%) and the speedup over LRU is marginal. By 30k timesteps the agent has discovered the sequential-growth structure of multi-turn conversations and prefetch accuracy approaches 90%. At convergence (50k timesteps) prefetch accuracy reaches 100% and the agent matches the Oracle speedup of 1.200×.

For the No-Context workload the prefetch accuracy remains near 0% throughout training, confirming the agent does not overfit spurious patterns when none exist. The RAG workload shows a non-monotonic trajectory: accuracy rises to ≈30% at 20k timesteps but then partially retreats as the agent learns to reduce wasted bandwidth by issuing fewer (but more selective) prefetches.

F. Discussion

Our evaluation reveals that RL-based prefetching is most effective when workload access patterns are *temporally predictable*. Multi-Turn conversations, which exhibit a strictly sequential growth pattern, allow the agent to learn a near-optimal deterministic policy. Shared Prefix workloads benefit from the agent confirming that the prefix has already been warmed up and avoiding redundant prefetches.

The RAG workload remains challenging due to the mismatch between the small training cache capacity and the number of unique document chunks per query. Increasing L2

TABLE III
HARDWARE EVALUATION RESULTS ACROSS ALL WORKLOADS

Workload	Strategy	Avg Lat. (ms)	Hit Rate (%)	Prefetch Acc. (%)	Total Prefetches	Speedup vs LRU
Prefix	No Cache	111.65	0.0	–	–	0.585×
	LRU	65.55	32.67	–	–	1.000×
	Oracle	65.33	32.67	–	–	1.003×
	RL Agent	**65.31**	32.67	0.0	218	**1.004**×
RAG	No Cache	300.98	0.0	–	–	0.167×
	LRU	50.34	0.0	–	–	1.000×
	Oracle	50.41	49.0	–	–	0.999×
	RL Agent	56.20	4.25	26.13	222	0.896×
No Context	No Cache	37.08	0.0	–	–	0.883×
	LRU	32.72	0.0	–	–	1.000×
	Oracle	32.60	0.0	–	–	1.004×
	RL Agent	32.74	0.0	0.0	180	0.999×
Multi-Turn	No Cache	981.60	0.0	–	–	0.108×
	LRU	105.79	2.20	–	–	1.000×
	Oracle	88.12	15.37	–	–	1.200×
	RL Agent	**88.30**	3.76	**100.0**	185	**1.198**×

TABLE IV
MULTI-TURN RESULTS AT DIFFERENT PPO TRAINING TIMESTEPS

Timesteps	Avg Lat. (ms)	Prefetch Acc. (%)	Speedup vs LRU
10,000	98.4	58.3	1.076×
20,000	93.1	74.6	1.136×
30,000	90.2	88.9	1.173×
40,000	89.1	95.7	1.187×
50,000	**88.3**	**100.0**	**1.198**×
Oracle	88.1	–	1.200×

cache capacity (from 4.5 MB to 48 MB, for example) could dramatically improve RAG hit rates.

The No-Context workload correctly yields no improvement, providing a sanity check that the agent does not erroneously prefetch in unpredictable scenarios.

G. Model Cost

Inference cost: The policy network (403→64→64→16, ≈30k parameters) executes entirely on CPU in under 1 ms per query, adding negligible per-request overhead. The asynchronous nature of L3→L2 prefetches (8.7 ms average) allows prefetch I/O to overlap with LLM prefill computation, further reducing effective latency impact.

Training cost: Behavioral cloning (20 epochs over 50 training samples) completes in under 30 s on a laptop CPU. PPO fine-tuning for 50,000 timesteps with real hardware I/O takes 10–30 minutes on a single RTX 3050 Laptop GPU. There are no cloud resources, large datasets, or expensive GPU clusters required—the entire pipeline runs on commodity hardware.

Storage cost: The trained policy checkpoint (`ppo_hardware_final.zip`) is under 500 KB. Trace CSV files and embeddings total approximately 2 MB per workload. The total software artefact footprint is under 10 MB.

H. Limitations

- **Small cache capacity**: The current L1 (3 MB) and L2 (4.5 MB) budgets are intentionally constrained to create cache pressure during training. In production, larger VRAM and DRAM allocations to KV cache would reduce eviction pressure but require re-tuning of the reward hyperparameters α, β, γ.
- **Fixed candidate set**: The action space considers only the top-16 most recently accessed L3 chunks as prefetch candidates. In environments with very large working sets this may exclude high-value cold chunks.
- **Single-model, single-node**: Neuro-Tiering is evaluated on a single GPU node serving one model. Extension to multi-GPU or multi-node deployments with shared KV caches (e.g., disaggregated prefill/decode) would require a distributed state representation.
- **Trace stationarity**: The PPO agent is trained and evaluated on traces from the same workload distribution. Significant workload drift (e.g., a sudden shift from multi-turn to RAG queries) may require online re-training or a mixture-of-experts policy.
- **RAG performance**: The agent currently underperforms LRU on RAG workloads due to limited L2 capacity relative to per-query document context size. This is an open problem requiring either larger cache budgets or a smarter candidate selection strategy.

I. Real-World Use Cases

Neuro-Tiering is directly applicable to any LLM serving scenario where the same KV cache chunks are accessed repeatedly across queries:

- **Multi-turn chatbots and assistants**: Customer support bots, coding assistants, and interactive tutors maintain growing conversation histories. The agent's near-Oracle performance on the Multi-Turn workload makes it immediately deployable in production chat services.

- **Document Q&A and RAG pipelines**: Enterprise search systems frequently query over a fixed document corpus. With sufficient L2 cache capacity the agent can pre-cache hot document chunks, reducing TTFT for repeated document retrievals.
- **Shared-prefix batch serving**: Applications that prepend long system prompts (e.g., legal or medical instruction templates) to every user query benefit from the agent confirming that shared prefix chunks are already warm and avoiding redundant prefetches.
- **Edge and resource-constrained inference**: The sub-1 ms policy inference cost and 500 KB model size make Neuro-Tiering viable on edge servers (e.g., NVIDIA Jetson) where VRAM is severely limited and tiered memory management is critical.
- **LMCache-integrated deployment**: Because Neuro-Tiering intercepts cache read/write events at the CacheEngine layer, it can be integrated into LMCache as a drop-in prefetch policy plugin without changes to vLLM or the model server.

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APPENDIX

A. Prerequisites

```
# Python virtual environment
python -m venv .venv
source .venv/bin/activate

# PyTorch with CUDA (RTX 3050)
pip install torch torchvision \
  --index-url \
  https://download.pytorch.org/whl/cu121

# Project dependencies
pip install -r requirements.txt

# Verify CUDA
python -c "import torch; \
  print(torch.cuda.is_available())"
```

B. Running the Full Pipeline

The pipeline consists of four stages executed in order:

Step 1 – Hardware Benchmark: Measures real L1/L2/L3 access latencies on your hardware. Output saved to `results/hardware_benchmark_latencies.csv`.

```
python -m env.hardware.benchmark
```

Step 2 – Training: Runs Stage A (behavioral cloning) followed by Stage B (PPO fine-tuning). Takes 10–30 minutes on an RTX 3050.

```
# Quick smoke test (< 2 min)
python train_hardware.py --quick

# Full training
python train_hardware.py

# With custom cache sizes
python train_hardware.py \
  --l1-mb 24 --l2-mb 32
```

Step 3 – Evaluation: Runs the trained RL agent and all baselines (LRU, No Cache, Oracle) on all four workloads. Results saved to `results/hardware_metrics.json` and `results/hardware_evaluation.csv`.

```
python eval/evaluate_hardware.py
```

Step 4 – Plotting: Generates all graphs from the results.

```
python eval/plot_results.py
```

C. Running the Convenience Shell Script

The full pipeline can be run with the provided script:

```
chmod +x run_pipeline.sh
./run_pipeline.sh
```

D. Directory Structure

```
rl_cache_prefetch/
+-- data/          Trace CSVs + embeddings
+-- env/           Cache simulator + Gym env
|   +-- hardware/  Real hardware backends
+-- agent/         Policy network + encoder
+-- eval/          Evaluation + baselines
+-- configs/       PPO + hardware configs
+-- models/        Saved model checkpoints
+-- results/       Metrics + plots
```

E. Configuration

Key parameters in `configs/hardware_config.yaml`:

- `l1_capacity_mb`: GPU VRAM budget for KV cache (default: 3 MB).
- `l2_capacity_mb`: CPU RAM pinned budget (default: 4.5 MB).
- `total_timesteps`: PPO training steps (default: 50,000).
- `behavioral_cloning_epochs`: Stage A epochs (default: 20).
- `alpha/beta/gamma_reward`: Reward function weights.

F. Expected Output

After a successful run, the following metrics are produced in `results/hardware_metrics.json`. Key expected results for the Multi-Turn workload:

- RL Agent: $\approx$88 ms avg latency (vs. 105 ms for LRU)
- Prefetch Accuracy: 100%
- Speedup vs LRU: 1.198$\times$